INCH-POUND

MIL-R-23417/12A
25 April 1991

SUPERSEDING
MIL-R-23417/12
15 September 1967

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 15R28W4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Electrical Resolvers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Excitation winding	Stator	---	---
Rated frequency	400	Hz	±1%
Input voltage:			
Maximum	26	volts (rms)	±1%
Test	10	volts (rms)	±1%
Minimum	0.5	volts (rms)	±1%
Primary current @ test voltage	4.0	milliamps	nominal
Primary power	0.006	watts	nominal
D.C. resistance:			
Rotor	200	ohms	±15%
Stator	295	ohms	±15%
Compensator	295	ohms	±15%
Impedance:			
Zro	470 +j2990	ohms	R±15%+jx±10%
Zso	540 +j2800	ohms	R±15%+jx±10%
Zrs	520 +j250	ohms	reference
Zco	540 +j2800	ohms	R±15%+jx±10%
Zcs	530 +j21	ohms	reference

AMSC - N/A FSC 5990

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TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
Resonant frequency:			
Rotor	60	kHz	minimum
Compensator	100	kHz	minimum
Low corner frequency	42	Hz	reference
Transformation ratio:			
Output/input	0.980	---	±1%
Rotor/compensator	0.985	---	±0.1%
Transformation ratio variation (rotor/ compensator) due to voltage with test voltage as reference	±0.05	percent	maximum
Equality of transformation ratio:			
Output/input	±0.08	percent	maximum
Rotor/compensator	±0.08	percent	maximum
Phase shift:			
Output/input	6.0	degrees	±1.0
Rotor/compensator	0	degrees	±0.20
Phase shift variation: (Rotor/compensator) due to voltage with test voltage as reference	±4.0	minutes	maximum
Interaxis error	3.0	minutes	maximum
Functional error	±0.1	percent	maximum
Null voltage @ test voltage:			
Fundamental	5.0	millivolts	maximum
Total	10	millivolts	maximum
Shift of resolver zero:			
With voltage	±1.0	minutes	maximum
With frequency	±1.0	minutes	maximum
Harmonic distortion	0.1	percent	maximum
Temperature rise	15	degrees C	maximum
Mechanical breakaway torque:	0.2	oz-in	maximum
@ -55°C	0.4	oz-in	maximum
Radial play	0.0005	inches	maximum
End play	0.0010	inches	±0.0005

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U ±.0000 -.0005
M23417/12-01A	.555	.500	2.100	Spline	Terminal Board	.1872
M23417/12-02A	.305	.250	1.850	Plain	Wire Leads	.1200
M23417/12-03A	.492	.437	1.850	Plain	Wire Leads	.1200
M23417/12-04A	.555	.500	1.850	Plain	Wire Leads	.1200
M23417/12-05A	.305	.250	2.100	Plain	Terminal Board	.1200
M23417/12-06A	.492	.437	2.100	Plain	Terminal Board	.1200
M23417/12-07A	.555	.500	2.100	Plain	Terminal Board	.1200

1/ Part identifying number M23417/12-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Part identifying number cross-reference. 1/

SUPERSEDED PART NUMBERS	NEW PART IDENTIFYING NUMBERS
MIL-R-23417/12	
M23417/12-01	M23417/12-01A
M23417/12-02	M23417/12-02A
M23417/12-03	M23417/12-03A
M23417/12-04	M23417/12-04A
M23417/12-05	M23417/12-05A
M23417/12-06	M23417/12-06A
M23417/12-07	M23417/12-07A

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

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NOTES:

1. Transformation ratio variation and phase shift variation (rotor/compensator) shall be tested as follows:

   a. Qualification: Variation shall remain within specification limits when subjected to individual or simultaneous variations of voltage (0.8 to 10 volts), frequency (380 to 420 Hz), and temperature (-55°C to +125°C). Test by applying input voltages of 0.8, 5.0, and 10 volts with frequency and temperature extremes simultaneously applied.

   b. Acceptance: Test by applying input voltages of 0.8, 5.0, and 10 volts at rated frequency and room ambient temperatures.

2. A voltage value of 0.5 to 26 volts shall be entered in the "volts" designation space on the identification plate.

Custodians:
Army - AR
Navy - AS
Air Force - 85

Review activities:
Navy - EC
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - MI, ME
Navy - OG, MC
Air Force - 19

Preparing activity:
Army - AR

(Project 5990-0381)

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